



Oral Surgery

Third stage

Lec. 16

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Elevators

- Are exo-levers, instrument designed to elevate or luxate the teeth or roots from their bony socket in close or surgical method of extraction to force a tooth or root along the line of withdrawal.



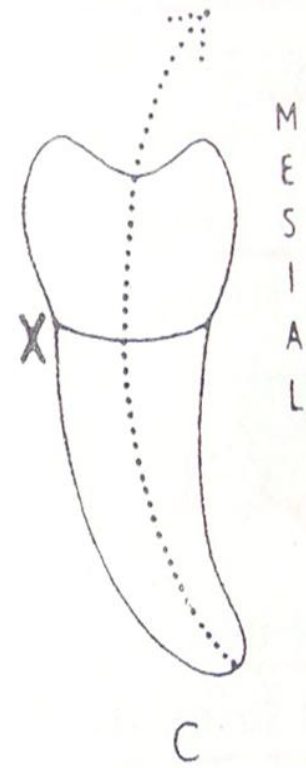
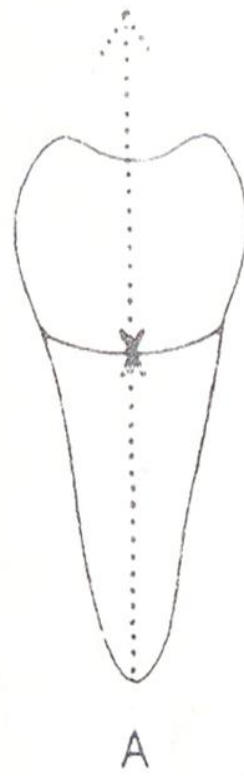
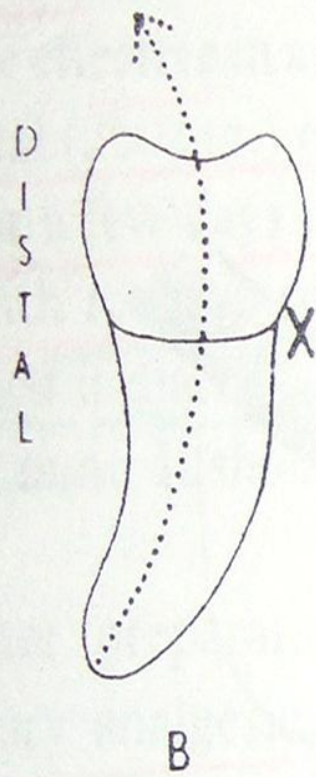
Elevators

- Line of withdrawal:-

Is the path along which the tooth or root will move out of its socket when minimal force is applied to it, and this line is primarily **determined by root pattern** (long axis of the tooth).

- Point of application:-

Is the site on the root at which force must be applied to effect delivery, it is determined by the line- of withdrawal. We have **buccal point of application, distal point of application, and mesial point of application.**



... with differing root

Blade Shank Handle



Parts of elevators:-

- All elevators have the following parts:-
- I-Handle: The handle of the elevator is usually of generous size, so it can be held comfortably in the hand to apply substantial, but controlled, force. The application of specifically applied force is critical in the proper use of dental elevators. This maybe a continuation of shank or at right angle to it.

Parts of elevators:-

- II- Shank. The shank of the elevator simply connects the handle to the working end, or blade, of the elevator. The shank is generally of substantial size and is strong enough to transmit the force from the handle to the blade.
- III- Blade: This part engages the crown or root and transmit force to the tooth, bone or both. The working side of the blade is either concave or flat. Always the concave surface of the working blade against the root or tooth.

Mechanical principles of using elevators:-

The work principles as applied to the use of elevators maybe that of:-

- 1- Lever principle.
- 2- Wedge principle.
- 3- Wheel and axle principle.
- 4- Combination of these principles.

Clinical uses of elevators:-

- 1- Elevators are used to luxate and remove teeth which cannot be engaged or grasped by the beaks of forceps (e.g. impacted teeth, malposed teeth), also badly carious teeth, and teeth with heavy filling.
- 2- To remove old roots and fractured roots and sectioned roots.
- 3- To loosen teeth prior to use of forceps.
- 4- To split teeth which have had grooves cut in them, as in separation of roots.
- 5- To remove small amounts of bone to create point of application for the beaks of forceps, or removal of interseptal bone.
- 6- Any tooth resisting normal extraction force by extracting forceps.

Elevators commonly used

There are so many elevators available
but few are widely used because of
their efficiency.

1-straight elevator:-

Elevator in which the blade, shank, and the handle are straight. The working blade or end **is blind and round**, there are many types and sizes of straight elevators.





2-Coupland's chisel (elevator):

It is similar to straight elevator but the working end is sharp and straight cut, **used for chiselling of bone to create point of application or to split of teeth**. It's of different sizes, size 1, size 2, size 3, depending on the width of the working end.



3-Cryer's elevators:-

In this type the working blades are sharp, pointed and triangular in shape just like a claw, forming an angle with the shank of the elevators.

These are pair instrument mesial and distal (right & left) designed to fit the root surface on mesial and distal surface. It's mostly used for removal of retained root of the lower molar and for elevation for impacted teeth after surgical exposure of the bifurcation of the tooth.



Left

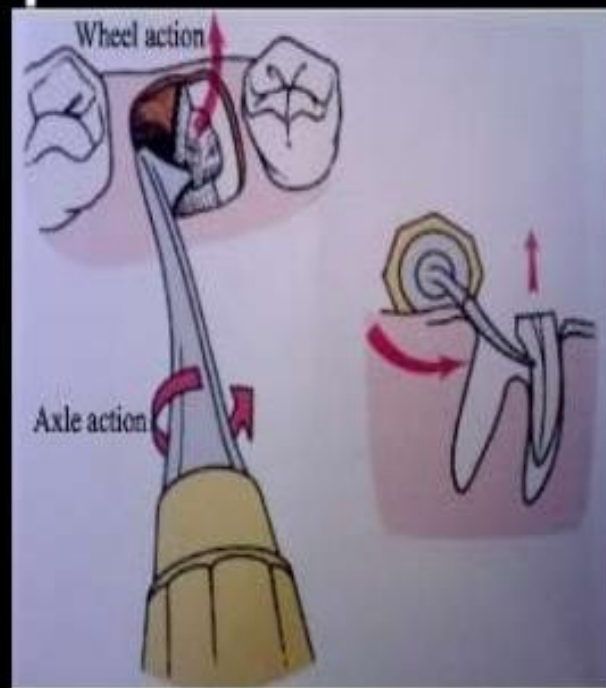


Right









4- Winter's elevator:-

In which the working end is the same that of Cryer's elevator but the handle is in right angle to the shank so it is called winter's (T-bar) cross-bar handle elevator. Winter's elevators are very powerful and great force may be applied or generated (sufficient to fracture the mandible) so the use of this elevator should be with great care to avoid fracture of the jaw.

Left







- 5- Apexo elevators:-
- The working blade is long, the margins are sharp, we have 3 Apexo, 2 angled and 1 straight (mesial, distal, straight).

- The blade forming an angle with the shank, this elevator is used mainly for removal of apical fragments of root deeply present in the socket of the lower jaw, especially molars. We push it between the socket and the root to loosen the fractured tip and remove it from the socket.



SUH01

SUH02

SUH03



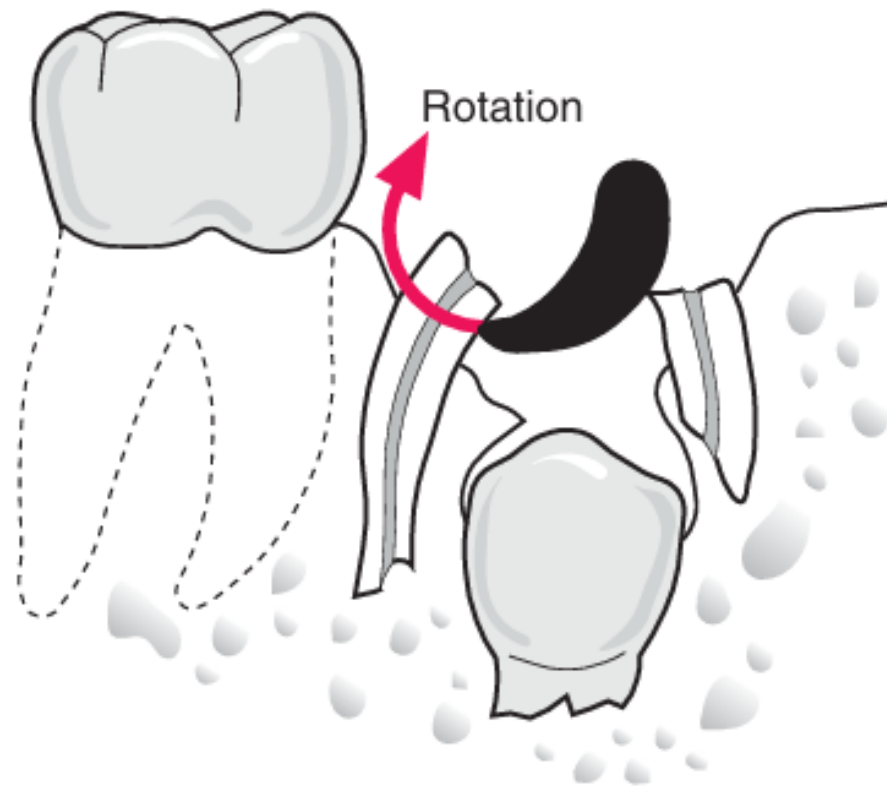
SUH2/3



- 6-War-wick James elevators:-

It is a light duty elevator. It's like Cryer's elevator, also we have two angled (mesial and distal) and one straight. The blade is short and the end is rounded and the handle is flattened, it's used for extraction of **retained roots, deciduous teeth, anterior lower teeth extraction, and where there is less resistance area. e.g. extraction of upper wisdom tooth.**





Removal of deciduous molar roots with a
Warwick James elevator



Position of an elevator in the palm of the hand.
Note the index finger along the length of the instrument.

Guiding principles for use of elevators:-

- The following rules should be observed when using elevators in general:-
- 1- Never use an adjacent tooth as a fulcrum, unless that tooth to be extracted itself in the same visit, and the fulcrum should always be bony one (alveolar bone).
- 2- An elevator should always be supported to avoid slippage and injury to the patient.
- 3- Avoid the use of excessive force if the tooth/root is resist luxation, by gentle rotation, then stop, look for the obstruction for elevation and deal with it.
- 4- The direction of force should be such that the roots are not directed toward major structures such as the maxillary antrum.

Guiding principles for use of elevators:-

- 5-An elevator should never be used “blindly” in the socket.
- 6- If an application point is not present, then this should be created by careful removal of bone.
- 7-Elevators should always be sterile and sharp.
- 8- The sharp edges of the working blades are placed between the alveolus and the root surface and gently rotated apically along the long axis of the elevator to luxate or displacing the tooth or root.

Complications of use of elevators:-

- Although elevators are very useful instrument for facilitating extraction of teeth, but misuse or miss-judgment may lead to some complications, part of it may be serious:-
- 1- Injury to the soft tissues, like injury to the tongue, floor of the mouth, soft and hard palate, caused by slipping of elevator during its use .
- 2- **Wrong application of force or excessive force** may lead to fracture of jaw especially the lower jaw at the angle of the mandible, also excessive force may lead to crushing of the alveolar bone and fragmentation
- 3- Fracture of maxillary tuberosity especially in extraction of upper third molars.

Complications of use of elevators:-

- 4- Uncontrolled force may lead to displacement of roots into maxillary sinus, infratemporal fossa, buccal soft tissue, submandibular space or inferior dental canal.
- 5- Use of elevator in periapical area of abscessed tooth may cause spread of infection to the surrounding tissue.
- 6- Tip of instrument (working blade) may be fractured and remain in the socket causing postoperative infection or delay healing, **so always check the tip of instrument after use.**

So most problems with elevators arise from:-

- a) Miss-judgement of amount of force exerted.
- b) Improper positioning of the elevators

thank
you

A stylized floral graphic featuring the words "thank" and "you" in a playful, hand-drawn font. The word "thank" is in brown with green outlines, and "you" is in brown with green outlines. The text is surrounded by various colorful flowers and leaves, including a red flower, an orange flower, a purple flower, a pink flower, and a white flower with a pink center. The overall style is whimsical and cheerful.